Taufique Shaikh

cs24mtech02006@iith.ac.in | staufique45@gmail.com | +91 8850826371 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.51
Jawahar Navodaya Vidyalaya, Raigad	Class XII (CBSE)	2017	82.80%
Jawahar Navodaya Vidyalaya, Raigad	Class X (CBSE)	2015	9.80%

M.TECH THESIS

RISC-V ISA Performance Characterization and Attribution

- Conducted instruction-level performance characterization of RISC-V ISA variants using the Spike simulator across 30 PolyBench/C kernels with four data types (int, float, double, long long).
- Analyzed instruction distributions across integer, memory, floating-point, control, vector, atomic, and system instruction classes to characterize workload behavior.
- Analyzed instruction-composition changes across transitions among 12 RV32/RV64 ISA configurations, identifying effects of data width, ISA extensions, compiler instruction selection, and memory operations.
- Investigated RISC-V Vector extension utilization.

Shared Memory Management in Heterogenous CPU-Accelerator System

- Modelled CPU-accelerator memory request flows in gem5-Aladdin to characterize memory interactions in heterogeneous systems.
- Implemented the DASH paper (deadline-aware DRAM scheduler) in gem5-Aladdin, analyzing CPU-accelerator memory interactions and shared DRAM behavior across heterogeneous systems.
- Modified gem5-Aladdin's memory management infrastructure to support deadline-aware DRAM request scheduling for multiple accelerators sharing memory resources.
- Characterized DRAM bandwidth utilization and memory access behavior across CPU-accelerator workloads.

PROJECTS

LLC Cache Replacement & Write Management | *ChampSim, C++, SPEC CPU*

- Implemented and evaluated LRU, SRRIP, SRRIP-HP, SRRIP-FP, DRRIP, and Random replacement policies at the LLC using SPEC CPU benchmarks.
- Implemented probabilistic write bypassing (10-30%) and analyzed its impact on LLC behavior and memory traffic.
- Developed LLC set-remapping and write-bypass mechanisms to study write distribution and cache utilization across workloads.

Work-Stealing Thread Pool | *C++20, Lock-Free Concurrency, Atomics, ThreadSanitizer, AddressSanitizer*

- Built a C++20 work-stealing thread pool using a lock-free Chase-Lev deque, with fork-join APIs, futures, and exception handling.
- Achieved up to 6.5× speedup on fine-grained recursive workloads and ~8× strong-scaling speedup on 20 cores on Ubuntu 20.04.
- Reduced per-task scheduling overhead up to 5.2× lower than a shared-queue baseline.
- Designed a lost-wake-up-free sleep/wake protocol and validated concurrency and memory safety using ThreadSanitizer, AddressSanitizer, and 2M-task stress tests.

CUDA SGEMM Optimization | *CUDA C++, cuBLAS, Tensor Cores (WMMA), Nsight Compute*

- Built 7 progressively optimized GEMM kernels in CUDA C++, going from 62 GFLOP/s to 3.67 TFLOP/s (59×), reaching 89% of cuBLAS FP32 throughput on an NVIDIA T4 (4096×4096).
- Applied memory-hierarchy optimizations: global-memory coalescing (9.6× speedup), shared-memory tiling, 1D/2D register blocking and vectorized float4 loads.
- Implemented a tensor-core kernel (WMMA, FP16 inputs with FP32 accumulation) reaching 22.3 TFLOP/s, 94% of cuBLAS FP16.
- Built a CUDA profiling workflow using CUDA events and Nsight Compute to measure throughput, occupancy, memory bandwidth, and bank conflicts.

EXPERIENCE

Research Assistant | Computer Architecture

- Analyzed instruction-level performance across RISC-V ISA configurations (RV32I, RV64IM, RV32G, RV64GV, etc.) using *spike + proxy kernel*.
- Performed detailed attribution for ISA extension and datapath width variation over curated workloads.
- Implemented DASH paper (Deadline-aware DRAM memory schedular) for heterogeneous CPU-accelerator systems using *gem5-Aladdin* and *C*.
- Extended *gem5-aladdin*'s memory management infrastructure to support deadline-aware DRAM request scheduling.

Member Technical | Full Stack Developer

- Developed end-to-end REST APIs using C# and ASP.NET
- Contributed to frontend development using React.js
- Implemented Selenium-based UI automation tests to validate application workflows and functional correctness.
- Performed manual end-to-end testing of deployed applications, validating REST APIs through Postman and UI functionality to verify workflows and identify defects.

TECHNICAL SKILLS

- **Programming:** C, C++, Python, Bash, SQL
- **Parallel Computing/HPC:** OpenMP, CUDA, Multithreading, SIMD
- **Computer Architecture/Systems:** RISC-V, Compiler, Multi-Core CPU, GPU, Pipelining, Cache Optimization, Shared-Memory Management
- **Simulation Tools:** Spike, gem5, gem5-aladdin, ChampSim
- **Development:** Git, Linux, CMake, Make, GDB, Docker, Vim
- **Libraries & Framework:** NumPy, Pandas, Matplotlib

RELEVANT COURSES

- **Computer Architecture:** Advanced Computer Architecture, Parallel Computing, Hardware Architecture for Deep Learning
- **Data Science:** Fraud Analytics, Multimedia Content Analysis, Linear Optimization
- **Systems:** Network Engineering, Operating Systems, DBMS

POSITIONS OF RESPONSIBILITY

- Placement Coordinator | Office of Career Services, IIT Hyderabad
- Coordinated national level workshop "Research Opportunities in Computer Architecture (ROCA 2025)".
- Core team member | Prakshepan - M.Tech orientation programme, IIT Hyderabad